

Assignment 3

(Due on 19th of May @11:59 PM)

The goal of this assignment is to experiment with optimization algorithms being utilized as multi-threshold detectors for gray scale images. Image segmentation quality is highly affected by both the number and values of the selected thresholds. Accordingly, this assignment is mainly aimed at finding the number and the values of multi-thresholds for a given image, given the gray wolf optimizer based on Otsu threshold detector. In other words, for a given image, multi-thresholds with different desired number of thresholds are to be calculated, aiming at finding the optimal number of thresholds.

You are expected to provide different experimentation scenarios alongside a comment on how is the optimal number of thresholds be decided.

You are asked to deliver the following:

1. Your function should take **grayscale** images and a set of **fixed thresholds** as input and return a segmented image based on these thresholds (**you must use grayscale images**).
2. A **notebook (.ipynb)** showing your implementation of both algorithms and helper functions, alongside test scenario results.
3. Prepare a detailed report analyzing the impact of different sets of threshold values on image segmentation using both algorithms. Your report should:
 - **Evaluate Threshold Values** – Experiment with a range of threshold values and document how they affect the segmentation process for the **3 images** provided below.
 - ◆ Image 1: http://tiny.cc/example_image1
 - ◆ Image 2: http://tiny.cc/example_image2
 - ◆ Image 3: http://tiny.cc/example_image3
 - **Determine the Most Suitable Thresholds** – Identify which threshold values yield the most effective segmentation results for each image and justify your selection, in other words, stating which set of thresholds produced the best visual results (provide comments on both the number of thresholds and their values and how does this affect the cohesiveness of the produced segments).

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→ **Output images generation**– the final output (of the best performing set of thresholds) should be presented in two ways:

- ◆ The conventional way: each segment is colored with the mean average of the pixels' values forming it.
 - ◆ The easy to visualize way: each segment is assigned a random RGB color.
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Submission Guidelines:

- This assignment must be completed in groups of **3–4 students**. All members must belong to tutorial groups supervised by the same **teaching assistant**. Each team must submit a .zip file with the following:
 - The **.ipynb** file containing your assignment's implementation (the notebook should be submitted showing the cells being run before and representing the output).
 - A shared link to the .ipynb file as well.
 - A report including all points mentioned earlier.
 - A .txt file with the team members' names and IDs.
 - The .zip file should have the following format: IP26_Assignment3.
 - Please use the following for submitting your assignment:

<https://forms.gle/y4Dg6Pt75bR5T4kFA>

Important Notes:

- Copying code or using external solutions without understanding is strictly prohibited.
- A plagiarism detection system will be used.
- Any violation will result in a grade of zero.
- Late submissions will not be allowed.